Minimax Methods in Geometric Analysis

Integrated lecture notes

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2026

These notes reorganize a four-part minicourse on minimax methods in geometric analysis. The presentation begins with finite-dimensional Morse theory, proceeds through the Palais deformation framework and viscous approximations, and ends with minmax constructions for the area of surfaces and the structure of their varifold limits.

Contents

Part I

Model Problems and Geometric Examples

1 The finite-dimensional model

The elementary saddle

$$f(x, y) = 1 + x^2 - y^2 \tag{1.1}$$

contains the basic geometry of a minimax construction. At the origin,

$$f(0,0) = 1, \qquad \nabla f(0,0) = 0, \qquad \nabla^2 f(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}. \tag{1.2}$$

Thus $\mathbb{R}^2 = E_2 \oplus E_{-2}$ and the Morse index of the critical point is one.

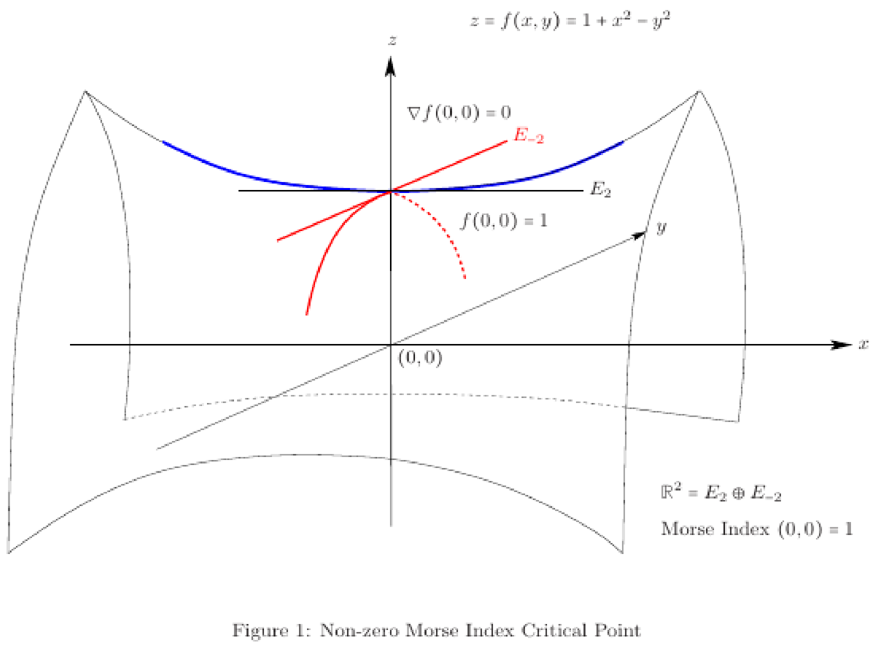

Figure 1: A nonzero-index critical point in the finite-dimensional model.

Set

$$\Omega_\pm = \{(x, y) \in \mathbb{R}^2 : f(x, y) \le 0, \ \pm y \ge 0\}. \tag{1.3}$$

The sublevel set $\{f \le 0\}$ has precisely two connected components. The admissible family is

$$\mathcal{A} = \left\{\gamma \in C^0([-1, 1], \mathbb{R}^2) : \gamma(\pm 1) \in \Omega_\pm\right\}. \tag{1.4}$$

If a homeomorphism Ψ of $\mathbb{R}^2$ is the identity on $\{f \le 0\}$, then $\Psi(\mathcal{A}) = \mathcal{A}$. Homologically,

$$\gamma \in \mathcal{A} \quad \Longleftrightarrow \quad [\gamma] \text{ generates } H_1(\mathbb{R}^2, \Omega_+ \cup \Omega_-; \mathbb{Z}) \simeq \mathbb{Z}. \tag{1.5}$$

This is the first example of a one-dimensional homological family.

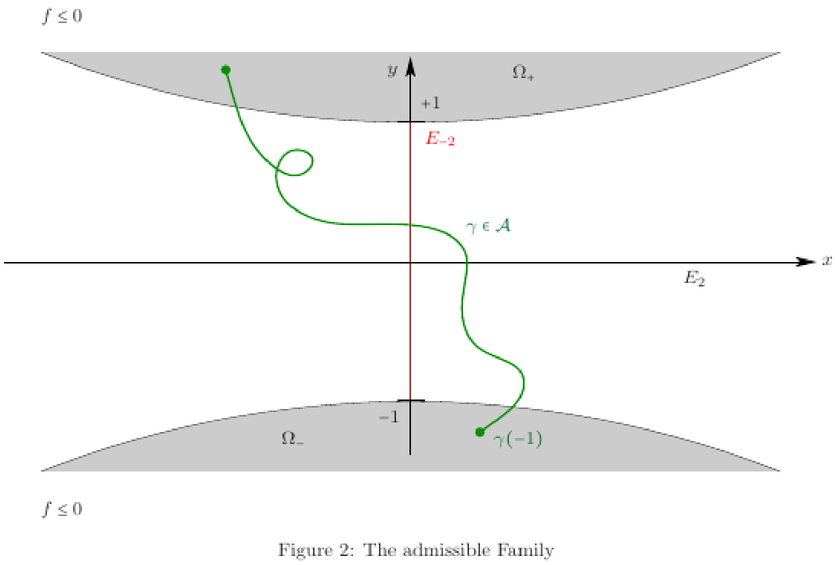

Figure 2: An admissible path joining the two components of the low sublevel set.

1.1 Width and pull-tight deformation

The width of the family is

$$W = \inf_{\gamma \in \mathcal{A}} \max_{s \in [-1,1]} f(\gamma(s)) = 1. \tag{1.6}$$

Indeed every admissible path intersects $\{y = 0\}$, where $f(x, 0) \geq 1$, and the unstable segment through the origin realizes the value one.

Introduce the descending vector field

$$X(x, y) = -\max\{f(x, y), 0\}\nabla f(x, y) \tag{1.7}$$

and its flow

$$\frac{\partial \Phi_t}{\partial t}(x, y) = X(\Phi_t(x, y)), \qquad \Phi_0(x, y) = (x, y). \tag{1.8}$$

Since the flow fixes $\{f \leq 0\}$, it preserves $\mathcal{A}$. The map $\gamma \mapsto \Phi_t \circ \gamma$ is the prototype of the *pull-tight* operation.

Proposition 1.1 (Realization of the width)**.** *Assume that* $|\nabla f|^{-1}([0, 1])$ *is compact. For every* $\varepsilon, \delta > 0$*, there exists* (x, y) *such that*

$$1 - \varepsilon < f(x, y) < 1 + \varepsilon, \qquad |\nabla f(x, y)| < \delta. \tag{1.9}$$

Consequently a convergent subsequence produces a critical point at level $W = 1$.

If, to the contrary, $|\nabla f| \geq \delta$ on $f^{-1}([1 - \varepsilon, 1 + \varepsilon])$, the flow would satisfy, for some $T > 0$,

$$\Phi_T\big(f^{-1}((-\infty, 1 + \varepsilon])\big) \subset f^{-1}((-\infty, 1 - \varepsilon)), \tag{1.10}$$

contradicting the definition of the width. Thus pull-tight becomes ineffective only near a critical point.

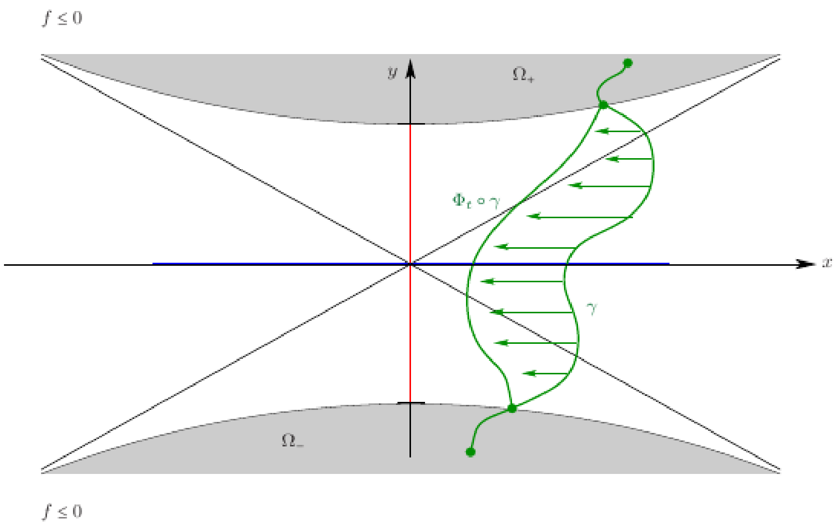

Figure 3: The pull Tight operation

Figure 3: The pull-tight deformation lowers a path away from critical points.

1.2 The higher-dimensional saddle

On $\mathbb{R}^{m+n}$, consider

$$f(x_1,\dots,x_m,y_1,\dots,y_n) = 1 + \sum_{i=1}^{m} x_i^2 - \sum_{j=1}^{n} y_j^2 \tag{1.11}$$

and let $\Omega = f^{-1}((-\infty,0])$. Since $\Omega \simeq S^{n-1}$, the long exact sequence of the pair gives

$$H_n(\mathbb{R}^{m+n},\Omega;\mathbb{Z}) \simeq H_{n-1}(\Omega;\mathbb{Z}) \simeq \mathbb{Z}. \tag{1.12}$$

An admissible family consists of maps

$$\gamma : X \longrightarrow \mathbb{R}^{m+n} \tag{1.13}$$

from an n-dimensional polyhedral chain such that $\gamma_*[X] \neq 0$ in $H_n(\mathbb{R}^{m+n},\Omega;\mathbb{Z})$.

Poincaré duality forces every admissible family to meet $\{y = 0\}$. Therefore

$$W = \inf_{\gamma\in\mathcal{A}} \max_{t\in X} f(\gamma(t)) \geq 1. \tag{1.14}$$

The unstable n-disc gives equality, and the width is achieved at the origin, whose Morse index is n.

2 Closed geodesics and Birkhoff sweep-outs

Let $N^n \subset \mathbb{R}^m$ be a closed submanifold and $u : S^1 \to N^n$. Its length and energy are

$$L(u) = \int_{S^1} |\partial_\theta u|\, d\theta, \qquad E(u) = \int_{S^1} |\partial_\theta u|^2\, d\theta. \tag{2.1}$$

For a variation u_s, with $w = \partial_s u_s|_{s=0}$, the first variation of length is

$$\frac{d}{ds} L(u_s)\bigg|_{s=0} Exitcode : 0Walltime : 2.4secondsOutput := \int_{S^1} \partial_\theta w \cdot \frac{\partial_\theta u}{|\partial_\theta u|}\, d\theta. \tag{2.2}$$

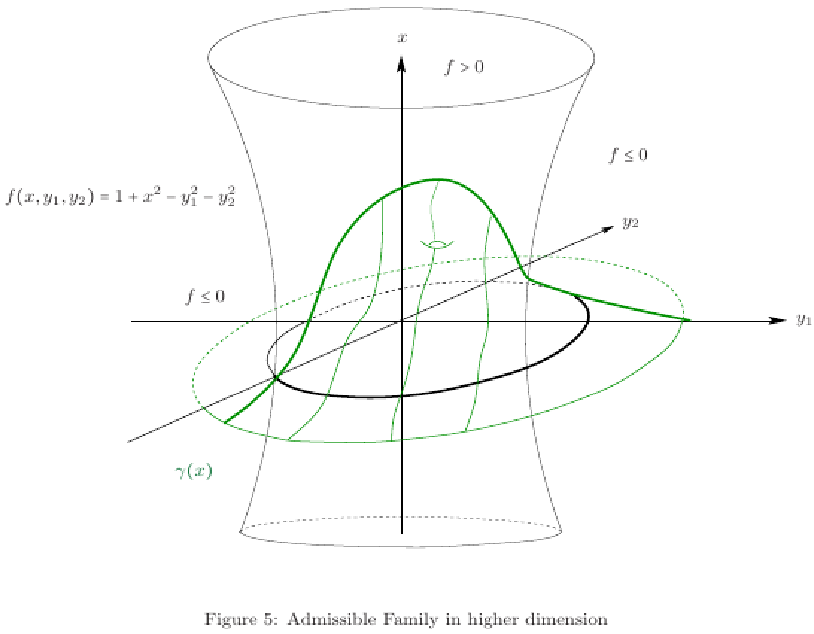

Figure 4: An admissible family for the higher-dimensional saddle.

In constant-speed parametrization this is equivalent to

$$P_u^T(\partial_{\theta\theta} u) = 0, \tag{2.3}$$

the geodesic equation. Equivalently, u is a critical point of the energy.

Theorem 2.1 (Hadamard–Poincaré–Cartan)**.** *If $\pi_1(N^n) \neq 0$, every nontrivial class $\alpha \in \pi_1(N^n)$ is represented by a closed geodesic.*

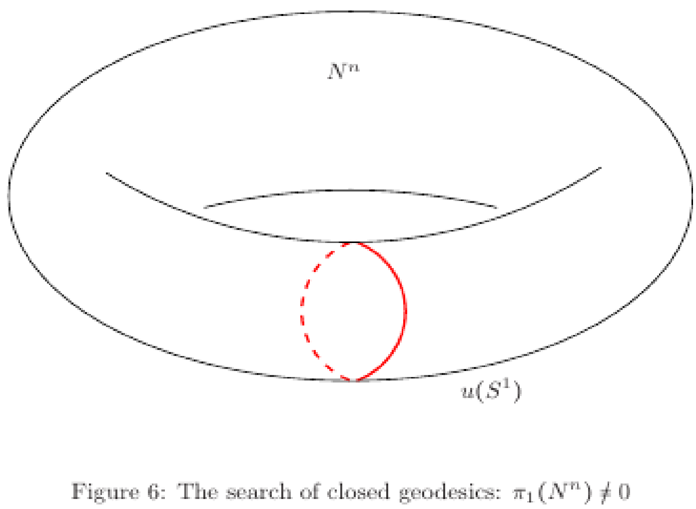

Figure 5: A nontrivial loop class represented by a closed geodesic.

To prove the theorem, minimize E in the homotopy class α. The inequalities

$$L(u)^2 \leq 2\pi E(u), \qquad |u(\theta) - u(\theta')| \leq |\theta - \theta'|^{1/2} E(u)^{1/2} \tag{2.4}$$

give compactness through the embedding $W^{1,2}(S^1) \hookrightarrow C^{0,1/2}(S^1)$. A minimizing sequence has a subsequence converging uniformly to u_∞. For sufficiently close loops, pointwise shortest geodesics give a homotopy, hence $[u_\infty] = \alpha$. The Euler–Lagrange equation is

$$\int_{S^1} \partial_\theta w \cdot \partial_\theta u_\infty \, d\theta = 0 \quad \text{for every } w \in T_{u_\infty} N. \tag{2.5}$$

It implies $P^T_{u_\infty}(\partial_{\theta\theta} u_\infty) = 0$ and $\partial_\theta u_\infty \cdot \partial_{\theta\theta} u_\infty = 0$. Thus the speed is constant and $L(u_\infty)^2 = 2\pi E(u_\infty)$.

2.1 The simply connected case

When $\pi_1(N^2) = 0$, minimizing a single loop produces a vanishing sequence. Birkhoff's remedy is to minimize over a family of loops.

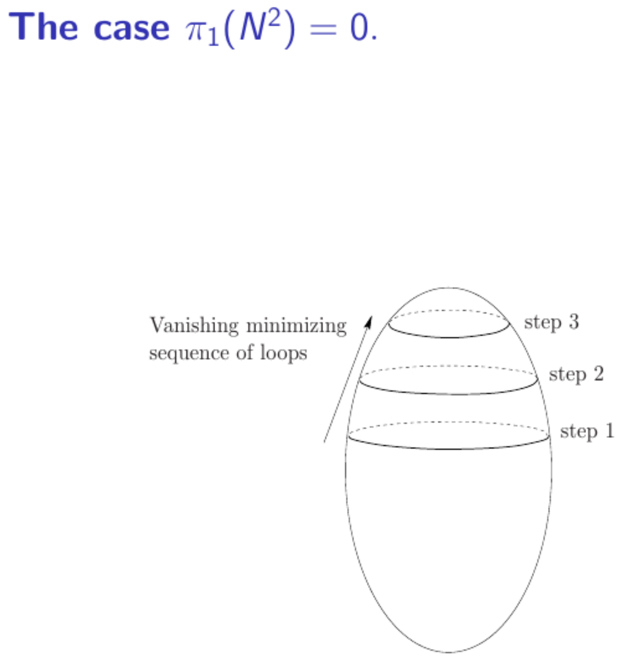

Figure 6: A minimizing sequence of contractible loops can collapse.

Definition 2.2 (Birkhoff sweep-out)**.** A sweep-out of a two-sphere N^2 is a map $u : [0,1] \times S^1 \to N^2$ such that

(i) $u \in C^0([0,1], W^{1,2}(S^1, N^2))$;

(ii) $u(0,\cdot)$ and $u(1,\cdot)$ are constant;

(iii) $u_*([0,1] \times S^1)$ generates $H_2(N^2; \mathbb{Z})$.

For the family $\mathcal{A}$ of sweep-outs define

$$W = \inf_{u \in \mathcal{A}} \max_{t \in [0,1]} E(u(t,\cdot)). \tag{2.6}$$

Lemma 2.3 (Nontriviality of the width)**.** *The Birkhoff width satisfies $W > 0$.*

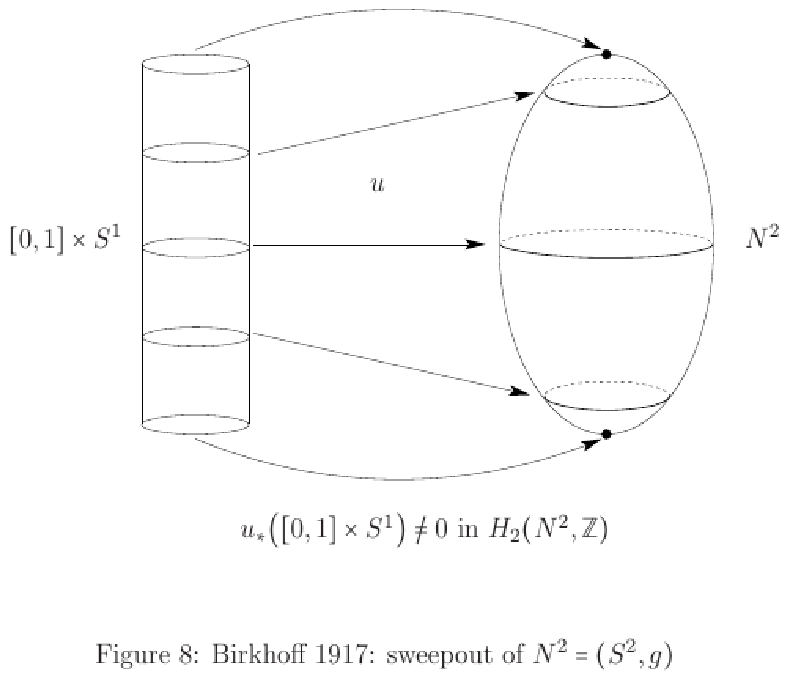

Figure 7: A Birkhoff sweep-out of (S^2, g).

If $W = 0$, a minimizing sequence u_k would satisfy

$$\max_t \operatorname{diam} u_k(t, S^1) \longrightarrow 0. \tag{2.7}$$

For large k, each loop lies in a convex geodesic ball centered at

$$p_k(t) = \pi_N \left(\fint_{S^1} u_k(t, \theta)\, d\theta \right). \tag{2.8}$$

Shortest geodesics contract every loop to $p_k(t)$, while the curve $p_k([0, 1])$ itself is contractible. This would make the sweep-out homologically trivial, a contradiction. Pull-tight then leads to the central question: is W realized by a constant-speed closed geodesic? The Palais theory developed later gives the affirmative answer.

3 The Plateau problem and minimal spheres

Let $D^2 \subset \mathbb{R}^2$ and let $u : D^2 \to \mathbb{R}^3$ span a Jordan curve Γ. The area and Dirichlet energy are

$$A(u) = \int_{D^2} |\partial_{x_1} u \wedge \partial_{x_2} u|\, dx^2, \tag{3.1}$$

$$E(u) = \frac{1}{2} \int_{D^2} |\nabla u|^2\, dx^2. \tag{3.2}$$

The pointwise arithmetic–geometric mean inequality gives

$$A(u) \leq E(u), \tag{3.3}$$

with equality exactly when u is weakly conformal:

$$|\partial_{x_1} u|^2 = |\partial_{x_2} u|^2, \qquad \partial_{x_1} u \cdot \partial_{x_2} u = 0. \tag{3.4}$$

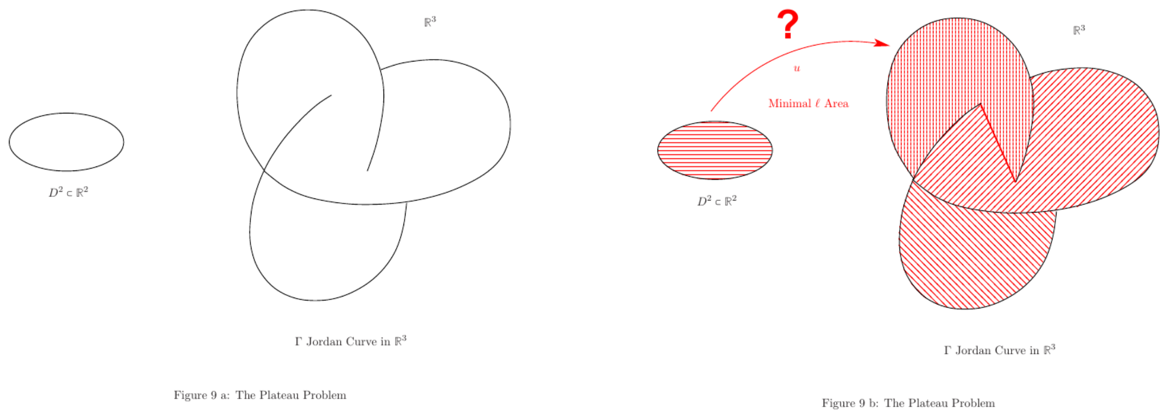

Figure 8: The Plateau problem: a Jordan curve and an area-minimizing spanning disc.

By uniformization, an immersion can be reparametrized conformally, so minimizing the more coercive functional E solves the area problem.

For a C^2 immersion, the first fundamental form is

$$g_u(X,Y) = u^* g_{\mathbb{R}^3}(X,Y) = \langle du(X), du(Y)\rangle. \tag{3.5}$$

If $\vec{n}_u : D^2 \to S^2$ is the Gauss map, the second fundamental form is

$$\vec{\mathrm{II}}_u(X,Y) = \langle d\vec{n}_u(X), du(Y)\rangle \vec{n}_u. \tag{3.6}$$

Its eigenvalues κ_1, κ_2 are the principal curvatures, and the mean-curvature vector is

$$\vec{H}_u = \frac{\kappa_1 + \kappa_2}{2}\vec{n}_u. \tag{3.7}$$

The first variation of area is

$$\frac{d}{dt}\operatorname{Area}(u + tw)\bigg|_{t=0} = -2\int_{D^2} \vec{H}_u \cdot w \; d\,\mathrm{vol}_{g_u}. \tag{3.8}$$

When u is conformal,

$$\vec{H}_u = \frac{1}{2}\frac{\Delta u}{|\partial_{x_1} u \wedge \partial_{x_2} u|}. \tag{3.9}$$

Hence a conformal immersion is minimal if and only if it is harmonic. The Douglas–Radó solution minimizes E among maps whose boundary parametrizes Γ monotonically; the minimizer is a conformal harmonic disc and therefore has least area.

3.1 Why direct minimization can lose topology

For maps $u : S^2 \to N^n$, set

$$A(u) = \int_{S^2} |du \wedge du|_{g_{S^2}} \; d\,\mathrm{vol}_{S^2}, \tag{3.10}$$

$$E(u) = \frac{1}{2}\int_{S^2} |du|^2_{g_{S^2}} \; d\,\mathrm{vol}_{S^2}. \tag{3.11}$$

Again $A(u) \le E(u)$, with equality for conformal maps, and critical points of E satisfy

$$P_u^T(\Delta_{S^2} u) = 0. \tag{3.12}$$

The difficulty is that $W^{1,2}(S^2)$ does not embed into C^0. The homotopy class is therefore not weakly closed at bounded energy. For instance, composing a degree-one map with dilating stereographic coordinates gives a sequence converging weakly to a constant while its energy concentrates at one point.

3.2 The Sacks–Uhlenbeck relaxation

For $\sigma > 0$, define

$$E_\sigma(u) = \int_{S^2} (1 + |du|^2)^{1+\sigma} \, d\,\mathrm{vol}_{S^2}. \tag{3.13}$$

The embedding

$$W^{1,2+2\sigma}(S^2) \hookrightarrow C^{0,\sigma/(1+\sigma)}(S^2) \hookrightarrow C^0(S^2) \tag{3.14}$$

is compact. Thus every class $\alpha \in \pi_2(N^n, x_0)$ has an E_σ-minimizer u_σ, satisfying

$$P^T_{u_\sigma} d^*\big((1 + |du_\sigma|^2)^\sigma du_\sigma\big) = 0. \tag{3.15}$$

Lemma 3.1 (Uniform small-energy regularity)**.** *There is $\varepsilon_N > 0$, independent of $0 \le \sigma \le 1$, such that*

$$\int_{B_r(x)} (1 + |du_\sigma|^2)^{1+\sigma} \, d\,\mathrm{vol}_{S^2} < \varepsilon_N \quad \Longrightarrow \quad |\nabla u_\sigma|(x) \le Cr^{-1}. \tag{3.16}$$

After passing to a subsequence, there are finitely many concentration points $a_1, \ldots, a_Q$ and a smooth harmonic map u_0 such that

$$u_{\sigma_k} \longrightarrow u_0 \quad \text{in } C^\ell_{\mathrm{loc}}(S^2 \setminus \{a_1, \ldots, a_Q\}) \quad \text{for every } \ell. \tag{3.17}$$

Point removability extends u_0 smoothly across the a_i. The original homotopy class need not be carried by u_0 alone: rescaling near the concentration points produces harmonic spheres.

Theorem 3.2 (Sacks–Uhlenbeck bubble tree)**.** *If $\pi_2(N^n) \neq 0$, a sequence of relaxed minimizers converges, in the bubble-tree sense, to finitely many nonconstant conformal harmonic maps*

$$u^1, \ldots, u^Q : S^2 \longrightarrow N^n. \tag{3.18}$$

Their images are minimal two-spheres, and together, up to the natural $\pi_1(N)$-action, they recover the original homotopy class.

3.3 Sweep-outs when $\pi_2(N) = 0$

For $N^3 \simeq S^3$, use sweep-outs by two-spheres:

$$\mathcal{A} = \left\{ u \in C^0([0,1], C^1(S^2, N^3)) : \begin{array}{l} E(u(0,\cdot)) = E(u(1,\cdot)) = 0, \\ u_*([0,1] \times S^2) \text{ generates } H_3(N^3; \mathbb{Z}) \end{array} \right\}. \tag{3.19}$$

Define

$$W = \inf_{u \in \mathcal{A}} \max_{t \in [0,1]} \frac{1}{2} \int_{S^2} |du(t,\cdot)|^2 \, d\,\mathrm{vol}_{S^2}. \tag{3.20}$$

The family is nonempty, admissible, and $W > 0$. Positivity follows from the small-energy harmonic map heat flow: sufficiently small two-spheres flow continuously to constants, so a small-energy sweep-out would be null-homologous.

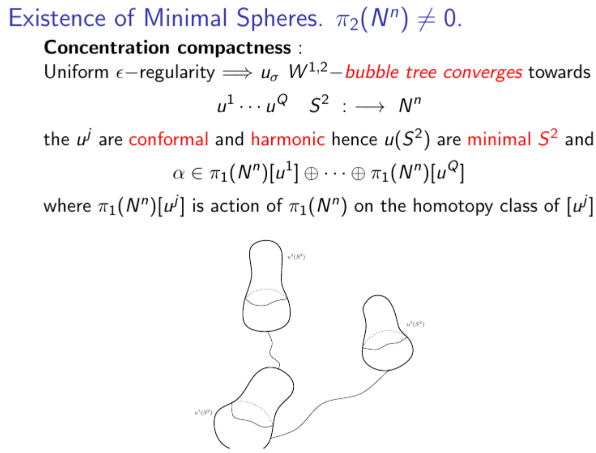

Figure 9: Bubble-tree formation in the Sacks–Uhlenbeck limit.

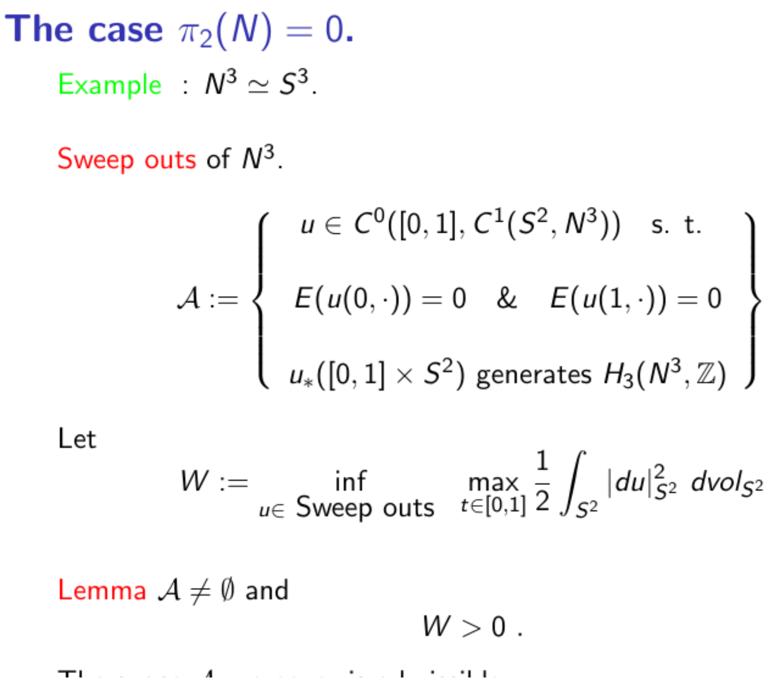

Figure 10: A sweep-out of a three-sphere by two-spheres.

4 Harmonic-map minmax problems

4.1 Mappings from a Riemann surface into a sphere

Let (Σ, g) be a closed oriented Riemannian surface. Consider families

$$u \in C^0_{L^2}(\overline{B^{n+1}}, W^{1,2}(\Sigma, S^n)) \tag{4.1}$$

whose boundary values are the constant maps $u(b,\cdot) \equiv b$ for $b \in \partial B^{n+1}$. Their width is

$$W = \inf_{u \in \mathcal{A}} \max_{a \in B^{n+1}} \frac{1}{2} \int_\Sigma |du(a,\cdot)|_g^2 \, d\,\mathrm{vol}_g \,. \tag{4.2}$$

The family is nonempty. One construction starts with an immersion $u_0 : \Sigma \to S^n$ and composes it with the conformal maps

$$\Phi_a(z) = (1 - |a|^2)\frac{z-a}{|z-a|^2} - a, \qquad a \in B^{n+1}, \tag{4.3}$$

setting $u(a,\cdot) = \Phi_{-a} \circ u_0$.

To prove $W > 0$, associate to a family the barycenter map

$$F_u(a) = \fint_\Sigma u(a,x)\; d\,\mathrm{vol}_g(x). \tag{4.4}$$

Since $F_u(b) = b$ on the boundary, degree theory gives an interior point a_0 with $F_u(a_0) = 0$. Poincaré's inequality yields

$$2W \ge \lambda_1(\Sigma, g)\,\mathrm{Area}(\Sigma, g). \tag{4.5}$$

Optimizing over the conformal class gives

$$2W \ge \Lambda_1(\Sigma, [g]), \tag{4.6}$$

the first conformal eigenvalue.

4.2 Maps between spheres

Let $n > 2$, $n \ge p$, and $\pi_n(S^p) \ne 0$. Fix a nontrivial map $v : S^n \to S^p$ and consider families

$$u : \overline{B^{n+1}} \longrightarrow W^{1,2}(S^n, S^p) \tag{4.7}$$

with constant boundary values $u(b,\cdot) \equiv v(b)$. A model family is $u(a,\cdot) = v \circ \Phi_{-a}$. Its width is

$$W = \inf_{u \in \mathcal{A}} \max_{a \in B^{n+1}} \frac{1}{2} \int_{S^n} |du(a,\cdot)|^2 \; d\,\mathrm{vol}_{S^n} \,. \tag{4.8}$$

If $W = 0$, Poincaré's inequality makes each map uniformly close in L^2 to its average. Projection of the average onto S^p would extend the nontrivial boundary map v across the ball, a contradiction. Hence $W > 0$.

For $p = n$, there is a parameter a_0 with zero average, and the sharp conformal estimate gives

$$W \ge \frac{n}{2}\,\mathrm{Area}(S^n). \tag{4.9}$$

In the case $n = 3$, $p = 2$, one expects an index-four harmonic map. A low-index classification identifies such a map, up to isometries and holomorphic reparametrization, with the Hopf fibration

$$h : S^3 \subset \mathbb{C}^2 \longrightarrow S^2 \subset \mathbb{R}^3, \qquad h(z,w) = \big(2z\overline{w}, |z|^2 - |w|^2\big). \tag{4.10}$$

This motivates the mapping form of the Willmore conjecture: the Hopf fibration should minimize the three-energy

$$E_3(u) = \int_{S^3} |du|^3 \, d\,\mathrm{vol}_{S^3} \tag{4.11}$$

among maps generating $\pi_3(S^2)$.

5 Willmore energy and sphere eversion

For a plane curve γ, Euler's elastica functional is

$$\mathcal{E}(\gamma) = \int_\gamma \kappa^2 \, ds. \tag{5.1}$$

Its surface analogue is the Willmore energy

$$W(u) = \int_\Sigma |\vec{H}_u|^2 \, d\,\mathrm{vol}_{g_u} \tag{5.2}$$

for an immersion $u : \Sigma \to \mathbb{R}^3$.

Theorem 5.1 (Willmore inequality)**.** *Every immersion of a closed oriented surface in $\mathbb{R}^3$ satisfies*

$$\int_\Sigma |\vec{H}_u|^2 \, d\,\mathrm{vol}_{g_u} \geq 4\pi. \tag{5.3}$$

Equality holds precisely for round spheres, up to conformal transformations of $\mathbb{R}^3 \cup \{\infty\}$.

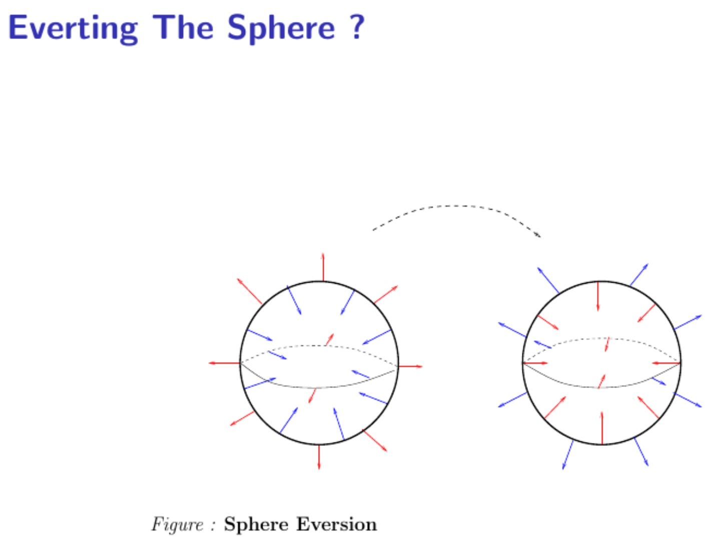

Figure 11: A sphere eversion connects the two orientations through immersions.

Let $\mathcal{A}$ be the family of regular homotopies from the standard sphere to the oppositely oriented one. Smale's theorem makes $\mathcal{A}$ nonempty. Define the eversion width

$$\beta = \inf_{u \in \mathcal{A}} \max_{t \in [0,1]} \int_{S^2} |\vec{H}_{u(t)}|^2 \, d\,\mathrm{vol}_{u(t)} \,. \tag{5.4}$$

Quantitative rigidity near the round sphere shows $\beta > 4\pi$: a path confined to a sufficiently small Willmore neighborhood of the round sphere cannot reverse the orientation.

The energy of a limiting critical immersion is quantized in multiples of 4π. The Li–Yau inequality implies that an immersion with a point of multiplicity k has Willmore energy at least $4\pi k$. These facts lead to the 16π conjecture for the optimal sphere eversion and to the expectation that the lowest saddle has a fourfold point at the singular symmetric stage.

Part II

Palais Deformation Theory

6 Banach manifolds and Finsler structures

Definition 6.1. A C^p Banach manifold $\mathcal{M}$ is a Hausdorff topological space covered by charts (U_i, φ_i), where $\varphi_i : U_i \to E_i$ takes values in Banach spaces and the transition maps are C^p.

The tangent spaces form a Banach vector bundle $T\mathcal{M} \xrightarrow{\pi} \mathcal{M}$. The basic global hypothesis is paracompactness: every open cover has a locally finite open refinement. Stone's theorem says that every metrizable space is paracompact. Consequently, a paracompact Banach manifold admits locally Lipschitz partitions of unity subordinate to arbitrary open covers. These partitions permit local constructions to be glued into global vector fields.

Definition 6.2. Let $\mathcal{V} \to \mathcal{M}$ be a Banach vector bundle over a normal Banach manifold. A Finsler structure is a continuous function

$$\|\cdot\| : \mathcal{V} \longrightarrow [0, \infty) \tag{6.1}$$

whose restriction to every fiber is a norm and whose local representatives are uniformly equivalent on compact subsets of a trivializing chart. A Banach manifold equipped with a Finsler structure on its tangent bundle is a Finsler manifold.

6.1 Sobolev immersions

Let Σ^2 be a closed oriented surface, let $N^n \subset \mathbb{R}^m$ be closed, and let $q > 2$. The space

$$\mathcal{M} = W^{2,q}_{\mathrm{imm}}(\Sigma, N) = \{\Phi \in W^{2,q}(\Sigma, N) : \operatorname{rank} d\Phi_x = 2 \text{ for every } x \in \Sigma\} \tag{6.2}$$

is a Banach manifold. Its tangent space at Φ is

$$T_\Phi \mathcal{M} = \{w \in W^{2,q}(\Sigma, \mathbb{R}^m) : w(x) \in T_{\Phi(x)} N\}. \tag{6.3}$$

Writing $g_\Phi = \Phi^* g_N$, one useful Finsler norm is

$$\|w\|_\Phi = \left(\int_\Sigma \left(|\nabla^2 w|^2_{g_\Phi} + |\nabla w|^2_{g_\Phi} + |w|^2 \right)^{q/2} d\,\mathrm{vol}_{g_\Phi} \right)^{1/q} + \left\| |\nabla w|_{g_\Phi} \right\|_{L^\infty(\Sigma)}. \tag{6.4}$$

It varies continuously with Φ and defines a Finsler structure.

6.2 The Palais distance

For $p, q \in \mathcal{M}$, let $\Omega_{p,q}$ be the set of C^1 paths joining p to q, and define

$$d_P(p,q) = \inf_{\omega \in \Omega_{p,q}} \int_0^1 \|\dot\omega(t)\|_{\omega(t)}\, dt. \tag{6.5}$$

Theorem 6.3 (Palais)**.** *The function d_P is a distance and induces the original topology of the Banach manifold. In particular, every Finsler manifold is metrizable and hence paracompact.*

For the norm above, $W^{2,q}_{\mathrm{imm}}(\Sigma, N)$ is complete with respect to d_P. The L^∞ control on $d\Phi$ keeps a Cauchy sequence away from loss of rank, while the integral terms give the required Sobolev convergence.

7 Pseudo-gradients and the Palais–Smale condition

Let $E \in C^1(\mathcal{M})$, and set

$$\mathcal{M}^* = \{u \in \mathcal{M} : DE_u \neq 0\}. \tag{7.1}$$

Definition 7.1. A pseudo-gradient for E is a locally Lipschitz section $X : \mathcal{M}^* \to T\mathcal{M}$ such that

$$\|X(u)\|_u < 2\|DE_u\|_u, \qquad DE_u[X(u)] > \|DE_u\|_u^2. \tag{7.2}$$

Local Hahn–Banach choices produce descent directions; a partition of unity glues them into a global pseudo-gradient. The associated flow is the infinite-dimensional pull-tight operation.

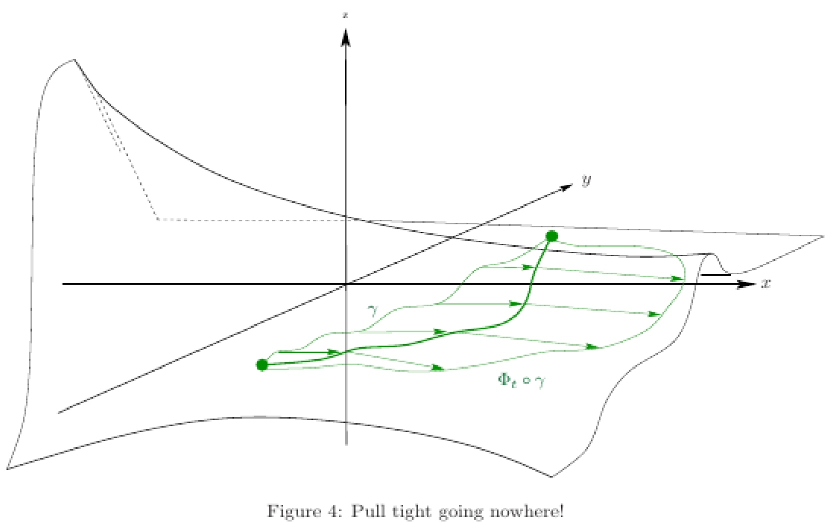

Figure 12: A pull-tight flow can stop only near a critical point.

Definition 7.2 (Palais–Smale condition)**.** The functional E satisfies $(\mathrm{PS})_\beta$ if every sequence $(u_k) \subset \mathcal{M}$ with

$$E(u_k) \longrightarrow \beta, Exitcode : 0Walltime : 2.2secondsOutput : \qquad \|DE_{u_k}\|_{u_k} \longrightarrow 0 \tag{7.3}$$

has a subsequence converging in d_P to some $u_\infty \in \mathcal{M}$. Necessarily $E(u_\infty) = \beta$ and $DE_{u_\infty} = 0$.

The energy on $W^{1,2}(S^1, N)$, equipped with its natural Sobolev Finsler norm, satisfies the Palais–Smale condition at every level. The compact embedding in one dimension is decisive here.

7.1 Admissible families

Definition 7.3. A collection $\mathcal{A}$ of closed subsets of $\mathcal{M}$ is admissible if every homeomorphism $\Psi : \mathcal{M} \to \mathcal{M}$ isotopic to the identity satisfies

$$A \in \mathcal{A} \quad \Longrightarrow \quad \Psi(A) \in \mathcal{A}. \tag{7.4}$$

Homotopy and homology classes of parameterized families give the standard examples. The sphere-eversion family, viewed in the immersion space modulo reparametrization, is one such admissible class.

Theorem 7.4 (Palais minmax principle)**.** *Let $(\mathcal{M}, \|\cdot\|)$ be a complete $C^{1,1}$ Finsler manifold, let $E \in C^1(\mathcal{M})$, and let $\mathcal{A}$ be admissible. Define*

$$\beta = \inf_{A\in\mathcal{A}} \sup_{u\in A} E(u). \tag{7.5}$$

If E satisfies $(\mathrm{PS})_\beta$*, then there is $u \in \mathcal{M}$ such that*

$$DE_u = 0, \qquad E(u) = \beta. \tag{7.6}$$

The proof is by contradiction. If no critical point lies at level β, the Palais–Smale condition supplies $\delta, \varepsilon > 0$ such that

$$|E(u) - \beta| < \varepsilon \quad \Longrightarrow \quad \|DE_u\|_u \geq \delta. \tag{7.7}$$

Choose a cutoff η supported near β and integrate

$$\frac{d}{dt}\phi_t(u) = -\eta(E(\phi_t(u)))X(\phi_t(u)). \tag{7.8}$$

Along this flow the energy decreases and

$$d_P(\phi_{t_1}(u), \phi_{t_2}(u)) \leq 2|t_2 - t_1|^{1/2}\big(E(\phi_{t_1}(u)) - E(\phi_{t_2}(u))\big)^{1/2}. \tag{7.9}$$

Completeness prevents finite-time escape. The time-one map is isotopic to the identity and deforms a near-optimal member of $\mathcal{A}$ below β, a contradiction.

8 Closed geodesics revisited

On $\mathcal{M} = W^{1,2}(S^1, N^2)$, the energy is Palais–Smale and the Birkhoff sweep-out class is admissible. The Palais theorem therefore shows that the positive Birkhoff width is achieved by a closed geodesic.

More generally, the free loop space

$$\mathcal{M} = W^{1,2}(S^1, M^m) \tag{8.1}$$

has the homotopy type of $C^0(S^1, M)$. Evaluation at a base point and the inclusion of constant loops split the homotopy exact sequence:

$$\pi_j(\mathcal{M}) \simeq \pi_j(\Omega_p M) \oplus \pi_j(M) \simeq \pi_{j+1}(M) \oplus \pi_j(M). \tag{8.2}$$

Suppose M is simply connected and $k \geq 2$ is the first integer with $\pi_k(M) \neq 0$. Then $\pi_{k-1}(\mathcal{M}) \simeq \pi_k(M) \neq 0$. Define

$$\mathcal{A}_k = \{u \in C^0(S^{k-1}, \mathcal{M}) : [u] \neq 0\}, \qquad W_k = \inf_{u\in\mathcal{A}_k} \max_{s\in S^{k-1}} E(u(s)). \tag{8.3}$$

Small loops can be contracted continuously, so $W_k > 0$. Palais minmax yields the following result.

Theorem 8.1 (Fet–Lyusternik)**.** *Every closed Riemannian manifold possesses a nonconstant closed geodesic.*

A geodesic is *prime* if it is not an iterate of a shorter geodesic. The existence of infinitely many prime closed geodesics is subtle and remains open for general metrics on S^n, $n \geq 3$.

9 The Gromov spectrum

For $\lambda > 0$, let

$$\mathcal{M}^\lambda = \{u \in W^{1,2}(S^1, M) : \sqrt{E(u)} \leq \lambda\}. \tag{9.1}$$

The Gromov dimension of this sublevel records the largest degree up to which relative homology vanishes:

$$\mathrm{dm}(\mathcal{M}^\lambda) = \sup\{k : H_j(\mathcal{M}, \mathcal{M}^\lambda; \mathbb{Z}) = 0 \text{ for every } j \leq k\}. \tag{9.2}$$

The nonlinear spectrum is then

$$\lambda_k = \sup\{\lambda > 0 : \mathrm{dm}(\mathcal{M}^\lambda) \leq k\}. \tag{9.3}$$

Theorem 9.1 (Gromov quasi-Weyl law)**.** *If $\pi_1(M)$ is finite, then*

$$\lambda_k \simeq k. \tag{9.4}$$

For a generic metric, Morse theory associates critical geodesics with homology generators. Combining this with the quasi-Weyl law gives lower bounds for the number of geodesics below a prescribed length. After removing iterates, one obtains bounds in terms of the Betti numbers of the loop space.

Theorem 9.2 (Gromoll–Meyer)**.** *Assume $\pi_1(M)$ is finite and*

$$\limsup_{k\to\infty} \dim H_k(\mathcal{M}; \mathbb{R}) = \infty. \tag{9.5}$$

Then every generic metric on M has infinitely many prime closed geodesics.

The minimal-model computation of Vigué-Poirrier and Sullivan shows that the homological hypothesis holds for a simply connected manifold whose real cohomology ring requires at least two generators. This does not cover spheres. In dimension two, however, the theorem of Franks and Bangert states that every Riemannian two-sphere has infinitely many prime closed geodesics.

Part III

Viscous Approximations of Minmax Operations

10 Why viscous regularizations are needed

The Dirichlet energy is critical in dimension two and therefore lacks the compactness needed for a direct Palais–Smale argument. A standard idea is to add a higher-order term. If (M^m, g) and (N^n, h) are closed oriented manifolds, a model functional is

$$E_\sigma(u) = \int_M |\nabla u|^2 \, d\,\mathrm{vol}_g + \sigma^2 \int_M |\nabla^\ell u|^p \, d\,\mathrm{vol}_g, \qquad \ell p > m. \tag{10.1}$$

The supercritical term gives compactness in $W^{1,p}$. If $N = F^{-1}(0) \subset \mathbb{R}^K$, another approximation is

$$\int_M |\nabla u|^2 \, d\,\mathrm{vol}_g + \frac{1}{\sigma} \int_M F(u)^2 \, d\,\mathrm{vol}_g \tag{10.2}$$

on the unconstrained space $W^{1,2}(M, \mathbb{R}^K)$.

The extra term solves the compactness problem, but its critical points may develop oscillations or concentration as $\sigma \to 0$. The length functional gives a transparent example.

11 A viscous approximation of length

Let $N^n \subset \mathbb{R}^m$ be closed and consider

$$\mathcal{M} = W^{2,2}_{\mathrm{imm}}(S^1, N^n). \tag{11.1}$$

For a curve parametrized by arclength, let $\vec{\kappa}_u$ denote its curvature and define

$$E_\sigma(u) = \int_{S^1} (1 + \sigma^2 |\vec{\kappa}_u|^2) \, d\ell_u. \tag{11.2}$$

The tangent norm

$$\|v\|_u = \left(\int_{S^1} (|\nabla^2 v|^2_{g_u} + |\nabla v|^2_{g_u} + |v|^2) \, d\ell_u \right)^{1/2} \tag{11.3}$$

turns $\mathcal{M}$ into a complete Finsler manifold, and E_σ is C^1. For fixed $\sigma > 0$, Palais minmax gives critical points modulo reparametrization:

$$E_\sigma(u_k) \to \beta(\sigma), \quad DE_\sigma(u_k) \to 0 \quad \Longrightarrow \quad u_{k'} \circ \psi_{k'} \to u \text{ in } d_P \tag{11.4}$$

for suitable $W^{2,2}$-diffeomorphisms $\psi_{k'}$ of S^1.

If $\mathcal{A}$ is admissible, write

$$\beta(\sigma) = \inf_{A \in \mathcal{A}} \sup_{u \in A} E_\sigma(u). \tag{11.5}$$

The minmax principle produces u_σ with $E_\sigma(u_\sigma) = \beta(\sigma)$ and $DE_\sigma(u_\sigma) = 0$. Weak convergence as $\sigma \to 0$ is not enough to identify a geodesic limit.

11.1 A first counterexample

Let $f(\sigma) = \sqrt{1 - 2\sigma^2}$ and set

$$u_\sigma(t) = \frac{\sigma}{f(\sigma)} \left(\cos \frac{f(\sigma)t}{\sigma}, \sin \frac{f(\sigma)t}{\sigma}, \frac{f(\sigma)}{\sigma} \sqrt{\frac{1 - 3\sigma^2}{1 - 2\sigma^2}} \right). \tag{11.6}$$

These are critical circles on S^2 in constant-speed parametrization. Their curvature and viscous energy satisfy

$$\kappa_{u_\sigma} \equiv -\frac{f(\sigma)}{\sigma}, \qquad E_\sigma(u_\sigma) = 2L(u_\sigma)(1 - \sigma^2) \longrightarrow \pi, \tag{11.7}$$

while

$$u_\sigma \longrightarrow (0,0,1), \qquad \sigma^2 \int_{S^1} \kappa^2_{u_\sigma} \, d\ell_{u_\sigma} \longrightarrow \frac{\pi}{2}. \tag{11.8}$$

The limiting map is constant, not a geodesic, and the derivatives do not converge strongly in L^1.

Consequently there is no $\varepsilon > 0$, independent of σ, for which

$$\int_{t_0 - r}^{t_0 + r} (1 + \sigma^2 \kappa^2_{u_\sigma}) \, d\ell_{u_\sigma} < \varepsilon \quad \Longrightarrow \quad |\ddot{u}_\sigma|(t_0) \le C r^{-1}. \tag{11.9}$$

A counter exemple

Precisely, let $f(\sigma) := \sqrt{1 - 2\sigma^2}$

$$u_\sigma(t) := \frac{\sigma}{f(\sigma)} \left(\cos\left(\frac{f(\sigma)}{\sigma} t \right), \sin\left(\frac{f(\sigma)}{\sigma} t \right), \frac{f(\sigma)}{\sigma} \sqrt{\frac{1 - 3\sigma^2}{1 - 2\sigma^2}} \right)$$

$$\lim_{\sigma \to 0} u_\sigma = (0, 0, 1)$$

$$\kappa_{u_\sigma}(t) \equiv -\frac{f(\sigma)}{\sigma}$$

$$E^\sigma(u_\sigma) = 2\, L(u_\sigma)\, (1 - \sigma^2) \;\longrightarrow\; \pi$$

In particular

$$\lim_{\sigma \to 0} \sigma^2 \int_{S^1} \kappa_{u_\sigma}^2 \; dl_{u_\sigma} = \frac{\pi}{2}$$

Figure 13: An explicit family of critical viscous circles.

Failure of $\epsilon-$Regularity

Conclusion : There is no $\epsilon-$regularity independent of σ. (Unlike Sacks Uhlenbeck relaxation).

That is $\nexists\, \varepsilon > 0$ s.t. in constant speed parametrisation

$$\int_{t_0 - r}^{t_0 + r} 1 + \sigma^2 \kappa_{u_\sigma}^2 \; dl_u < \varepsilon \quad \Longrightarrow \quad |\ddot{u}_\sigma|(t_0) \leq C\; r^{-1}$$

Figure 14: The viscous length approximation has no σ-uniform ε-regularity.

11.2 A second obstruction: Hopf degree

Let $u_\sigma : S^3 \to S^2$ minimize

$$H_\sigma = \min\left\{\int_{S^3} (|du|^2 + \sigma^2 |du|^4)\ d\,\mathrm{vol}_{S^3} : \text{Hopf-deg}(u) = 1\right\}. \tag{11.10}$$

One can have $H_\sigma \to 0$, hence $du_\sigma \to 0$ in L^2, while topology forces

$$\liminf_{\sigma\to 0} \int_{S^3} |du_\sigma|^3\ d\,\mathrm{vol}_{S^3} > 0. \tag{11.11}$$

Thus an energy bound cannot yield a uniform gradient estimate. The missing ingredient is a suitable monotonicity formula.

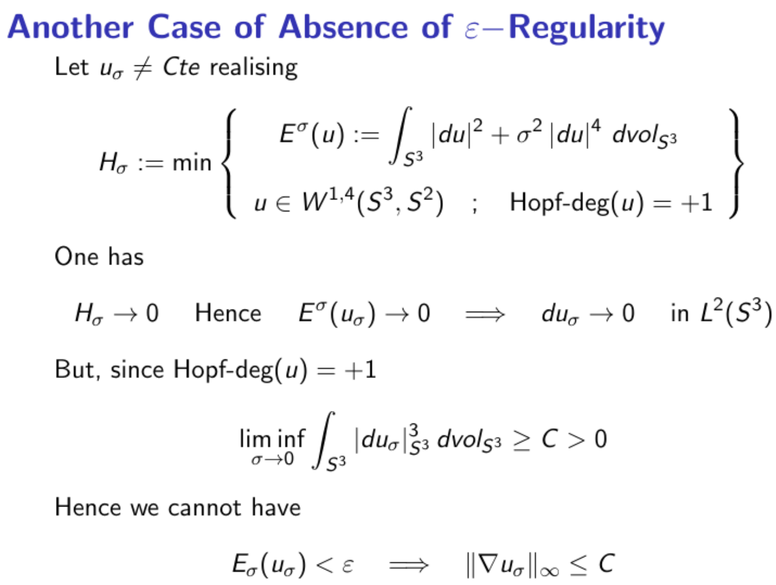

Figure 15: A topological obstruction to uniform regularity in the Hopf class.

12 Vanishing viscous energy

The failure above disappears under an additional hypothesis on the viscous part.

Theorem 12.1 (Vanishing viscous energy)**.** *Let $u_\sigma : S^1 \to N$ be critical points of E_σ, in constant-speed parametrization. Assume*

$$\limsup_{\sigma\to 0} L(u_\sigma) < \infty, \qquad \int_{S^1} \sigma^2 |\vec{\kappa}_{u_\sigma}|^2\, d\ell_{u_\sigma} \longrightarrow 0. \tag{12.1}$$

Then there are $\sigma_j \to 0$ and a geodesic u_0 such that

$$\dot{u}_{\sigma_j} \longrightarrow \dot{u}_0 \quad \textit{strongly in } L^1(S^1). \tag{12.2}$$

Here is the compactness mechanism. Write $|\dot{u}_{\sigma_j}| \equiv L_j/(2\pi)$, set $u_j = u_{\sigma_j}$, and use the covariant derivative $D_t = P^T_{u_j}\, d/dt$. The Euler–Lagrange equation can be written

$$D_t\left[\dot{u}_j - \sigma_j^2\left(2D_t^2 \dot{u}_j + 3\kappa_j^2 \dot{u}_j\right)\right] + 2\sigma_j^2 R(D_t \dot{u}_j, \dot{u}_j)\dot{u}_j = 0. \tag{12.3}$$

Define

$$w_j = \dot{u}_j - \sigma_j^2\big(2D_t^2\dot{u}_j + 3\kappa_j^2\dot{u}_j\big). \tag{12.4}$$

The equation and the vanishing viscous energy imply $D_t w_j \to 0$ in L^2. In local coordinates the Christoffel terms are controlled in $L^1 + H^{-1}$, so (w_j) is precompact in L^2. Moreover $w_j \to \dot{u}_0$ in distributions and

$$\int_{S^1} w_j \cdot \dot{u}_j \, dt = \int_{S^1} |\dot{u}_j|^2 \, dt + o(1) \longrightarrow \int_{S^1} |\dot{u}_0|^2 \, dt. \tag{12.5}$$

Uniform convexity then gives strong convergence of $\dot{u}_j$ in every L^p, $p < \infty$, and in particular in L^1. The weak Euler–Lagrange equation identifies u_0 as a geodesic.

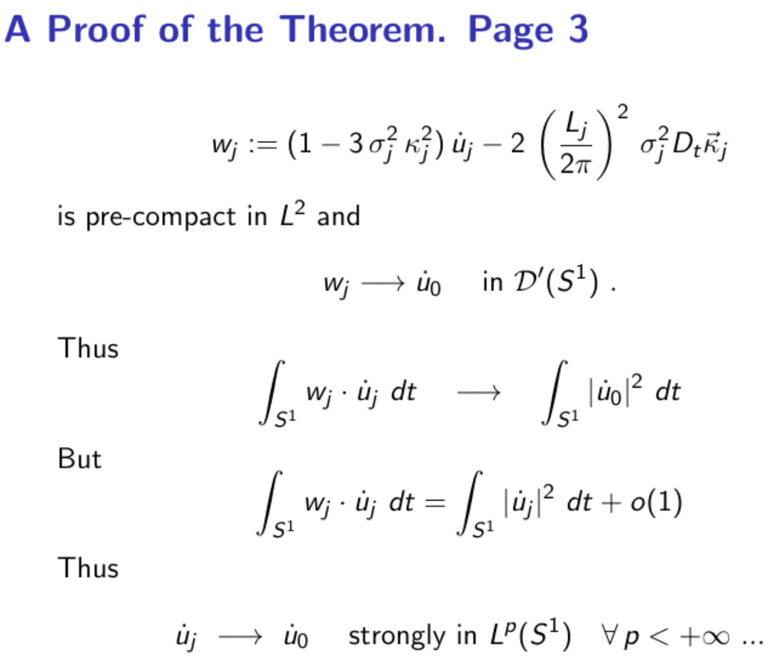

Figure 16: The compactness argument for the vanishing-viscous-energy theorem.

13 Struwe's monotonicity trick

Let $E_\sigma \in C^1(\mathcal{M})$, $\sigma \in [0,1]$, be increasing in σ, and assume that the derivatives vary continuously in the Finsler norm. Let $\mathcal{A}$ be admissible and

$$\beta(\sigma) = \inf_{A\in\mathcal{A}} \sup_{\Phi\in A} E_\sigma(\Phi). \tag{13.1}$$

Theorem 13.1 (Struwe)**.** *Assume the family E_σ satisfies the Palais–Smale condition. There are $\sigma_j \to 0$ and critical points Φ_j such that*

$$E_{\sigma_j}(\Phi_j) = \beta(\sigma_j), \qquad DE_{\sigma_j}(\Phi_j) = 0, \tag{13.2}$$

$$\partial_\sigma E_{\sigma_j}(\Phi_j) = o\left(\frac{1}{\sigma_j \log(1/\sigma_j)}\right). \tag{13.3}$$

The proof uses the monotonicity of β. Since β is monotone, it is differentiable almost everywhere and

$$\beta'(\sigma)\, d\sigma + d\mu = d\beta(\sigma), \qquad d\mu \perp d\sigma. \tag{13.4}$$

The integrability of β' near zero produces a sequence $\sigma_j \to 0$ with

$$\beta'(\sigma_j) = o\left(\frac{1}{\sigma_j \log(1/\sigma_j)}\right). \tag{13.5}$$

At a differentiability point, compare $\beta(\tau)$ with $\beta(\sigma)$ for $\sigma < \tau < \sigma + \delta$. A cutoff modifies the pseudo-gradient at parameter τ so that the flow is active only when

$$E_\sigma(\Phi) \geq \beta(\sigma) - \varepsilon(\tau - \sigma). \tag{13.6}$$

If the derivative were bounded away from zero on all near-optimal families, this flow would lower their E_σ-maximum below $\beta(\sigma)$, contradicting the definition of the width.

The Proof of Struwe Monotonicity Trick - page 1

$$\beta(\sigma) \searrow \beta(0) \quad \Longrightarrow \quad \beta \text{ is diff. a.e.}$$

and

$$D\beta(\sigma) = \beta'(\sigma)\ d\mathcal{L}^1 \llcorner [0,1] + \mu \quad \text{where} \quad \mu \perp d\mathcal{L}^1 \llcorner [0,1]$$

$$\int_0^\sigma \beta'(s)\ ds \leq \beta(\sigma) - \beta(0)$$

Hence $\exists\ \sigma_j \to 0$

$$\beta'(\sigma_j) = o\left(\frac{1}{\sigma_j\ \log \sigma_j^{-1}}\right)$$

Figure 17: The monotonicity estimate selects parameters with small viscous derivative.

For the viscous length, take

$$E_\sigma(\Phi) = L(\Phi(S^1)) + \sigma^2 \int_{S^1} |\vec{\kappa}_\Phi|^2\, d\ell_\Phi. \tag{13.7}$$

Struwe's theorem gives $\sigma_j \to 0$ and critical points Φ_{σ_j} with

$$\sigma_j^2 \int_{S^1} |\vec{\kappa}_{\Phi_{\sigma_j}}|^2\, d\ell = o\left(\frac{1}{\log(1/\sigma_j)}\right). \tag{13.8}$$

The vanishing-energy theorem then yields a geodesic Φ_0 with $L(\Phi_0) = \beta(0)$, recovering Birkhoff's result through viscous Palais–Smale approximations.

Part IV

Minmax Operations for the Area

14 The area minmax landscape

Let $(M^m, g) \hookrightarrow \mathbb{R}^n$ be a closed Riemannian manifold and let Σ^2 be a closed oriented surface. The configuration space is

$$\mathcal{M} = \{u : \Sigma^2 \to M^m :\ u \text{ is an immersion}\}. \tag{14.1}$$

An admissible family $\mathcal{A} \subset \mathcal{P}(\mathcal{M})$ is invariant under homeomorphisms of $\mathcal{M}$ isotopic to the identity. Its area width is

$$W_{\mathcal{A}} = \inf_{A\in\mathcal{A}} \sup_{u\in A} \operatorname{Area}(u). \tag{14.2}$$

Theorem 14.1 (Minmax for the area)**.** *Let $\mathcal{A}$ be an admissible n-homological family with $W_{\mathcal{A}} > 0$. Then there are a closed surface S, a smooth minimal branched immersion $v : S \to M$, and a multiplicity function $N \in L^\infty(S, \mathbb{N})$ such that*

$$\operatorname{genus}(S) \le \operatorname{genus}(\Sigma), \qquad \operatorname{Area}(v) = W_{\mathcal{A}}, \qquad \operatorname{index}(v) \le n. \tag{14.3}$$

The strategy is to preserve reparametrization invariance, perturb the area just enough to recover Palais–Smale compactness, apply the Palais deformation theory at fixed viscosity, and then pass to the limit. This avoids both the loss of conformal class in a direct Sacks–Uhlenbeck parameterization and the unresolved regularity issues of a purely geometric-measure-theoretic minmax.

15 A viscosity for the area

For an immersion u, let $\vec{n}_u : \Sigma \to G_{n-2}(\mathbb{R}^n)$ be its Gauss map. For $p > 1$, define

$$F^p_\sigma(u) = \operatorname{Area}(u) + \sigma^2 \int_\Sigma (1 + |d\vec{n}_u|^2)^p \; d\operatorname{vol}_{g_u} \tag{15.1}$$

on

$$\mathcal{M} = W^{2,2p}_{\mathrm{imm}}(\Sigma^2, M^m). \tag{15.2}$$

The functional is invariant under reparametrization and is coercive in the second derivatives of u.

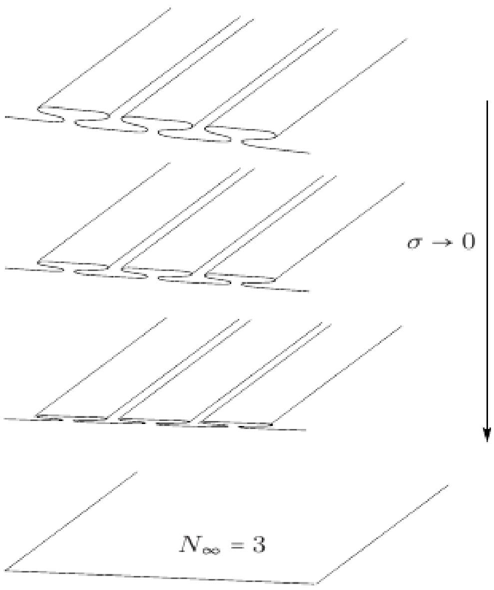

Figure 11: Weak Convergence in the Viscosity Method

Figure 18: Weak convergence in the viscosity method, with multiplicity in the limit.

15.1 Closure of the immersion space

Theorem 15.1 (Langer–Breuning compactness)**.** *Let $p > 2$, and let $u_k \in W^{2,p}_{\mathrm{imm}}(\Sigma, M)$ satisfy*

$$\limsup_{k\to\infty} \int_\Sigma \left(1 + |d\vec{n}_k|^2_{g_{u_k}}\right)^p \, d\,\mathrm{vol}_{g_{u_k}} < \infty. \tag{15.3}$$

Then there are a subsequence and diffeomorphisms $\varphi_k \in W^{2,p}_{\mathrm{diff}}(\Sigma, \Sigma)$ such that

$$v_k = u_k \circ \varphi_k \rightharpoonup v_\infty \quad \textit{weakly in } W^{2,p}(\Sigma, M), \tag{15.4}$$

and v_∞ is still an immersion.

The diffeomorphisms are essential: the energy controls the geometric surface, not a preferred parameterization.

15.2 Step 1: control of the conformal class

Assume genus$(\Sigma) > 0$. Write

$$g_{u_k} = e^{2\alpha_k} h_k, \tag{15.5}$$

where h_k is the volume-one constant-curvature metric in the conformal class of g_{u_k}. The first compactness step is

$$h_k \longrightarrow h_\infty \quad \text{in every } C^\ell \text{ topology} \tag{15.6}$$

after pull-back by diffeomorphisms.

For the torus, write the conformal domain as $\mathbb{R}^2/(\mathbb{Z} \times \tau_k\mathbb{Z})$, $\tau_k \geq 1$, and let

$$u_k^* g_{\mathbb{R}^n} = e^{2\alpha_k}(dx_1^2 + dx_2^2), \qquad e_k = e^{-\alpha_k}\partial_{x_1} u_k, \quad f_k = e^{-\alpha_k}\partial_{x_2} u_k. \tag{15.7}$$

Fenchel's inequality, followed by the Liouville equation

$$-\Delta\alpha_k = e^{2\alpha_k} K_k, \qquad |K_k| \leq |d\vec{n}_k|^2_{g_{u_k}}, \tag{15.8}$$

controls the conformal modulus:

$$\tau_k \leq C_p \frac{p}{p-1} \left(\int_\Sigma |d\vec{n}_k|^{2p}_{g_{u_k}} \, d\,\mathrm{vol}_{g_{u_k}} \right)^{1/p} \left(\int_\Sigma d\,\mathrm{vol}_{g_{u_k}} \right)^{1-1/p}. \tag{15.9}$$

Thus a degenerating sequence of conformal classes would force the viscous energy to diverge.

15.3 Step 2: control of the conformal factor

Assume now $h_k \to h_\infty$ smoothly. The Willmore inequality and Hölder's inequality imply

$$0 < c \leq \liminf_k \int_\Sigma e^{2\alpha_k} \, d\,\mathrm{vol}_{h_k} \leq \limsup_k \int_\Sigma e^{2\alpha_k} \, d\,\mathrm{vol}_{h_k} < \infty. \tag{15.10}$$

The Liouville equation becomes

$$-\Delta_{h_k}\alpha_k = e^{2\alpha_k} K_{g_{u_k}} - K_{h_k}. \tag{15.11}$$

Testing with $e^{-2s\alpha_k}$, $s = 1 - 1/p$, gives uniform L^q bounds for $e^{-\alpha_k}$, hence for α_k after using the weak L^2 estimate for $d\alpha_k$.

Small-energy disks are controlled by Hélein's estimate:

$$\int_{B_r^{h_k}(x_0)} |d\vec{n}_k|^2_{h_k}\, d\,\mathrm{vol}_{h_k} < \varepsilon \quad \Longrightarrow \quad \|\alpha_k - c_k\|_{L^\infty(B_{r/2})} \le C\big(1 + \|d\alpha_k\|_{L^{2,\infty}(B_r)}\big). \tag{15.12}$$

If concentration occurs, an ε-neck has the form

$$A(\eta_k, \delta_k) = B_{\eta_k}(x_k) \setminus B_{\delta_k/\eta_k}(x_k), \qquad \frac{\delta_k}{\eta_k^2} \to 0, \tag{15.13}$$

and Bernard–Rivière's estimate gives

$$\big\|\alpha_k - d_k \log d_{h_k}(x, x_k) - A_k\big\|_{L^\infty(A(\eta_k/2,\delta_k))} \le C\big(1 + \|d\alpha_k\|_{L^{2,\infty}}\big). \tag{15.14}$$

Combining a bubble with its neck would force $\int e^{2\alpha_k}\, d\,\mathrm{vol}_{h_k} \to \infty$, contradicting the area bound. Hence bubbles are excluded at this stage and

$$\limsup_{k\to\infty} \|\alpha_k\|_{L^\infty(\Sigma)} < \infty. \tag{15.15}$$

16 Euler–Lagrange equation and fixed-viscosity minmax

For a critical point of

$$F_\sigma(u) = \int_\Sigma f_\sigma(|d\vec{n}_u|^2)\, d\,\mathrm{vol}_{g_u}, \tag{16.1}$$

the Euler–Lagrange equation in $M = \mathbb{R}^3$ is

$$d^*_{g_u}\left[f_\sigma\, du - 2f'_\sigma\, d\vec{n} \,\dot\otimes\, d\vec{n} \cdot du + 2d^*_{g_u}(f'_\sigma d\vec{n}) \cdot du\, \vec{n}\right] = 0, \tag{16.2}$$

where $f_\sigma = f_\sigma(|d\vec{n}|^2)$. In conformal coordinates $g_u = e^{2\lambda}(dx_1^2 + dx_2^2)$, this becomes a divergence-form system. For the model functional, $f_\sigma(y) = 1 + \sigma^2 y^2$ and $f'_\sigma(y) = 2\sigma^2 y$.

Theorem 16.1 (Kuwert–Lamm–Li)**.** *For $p > 1$, the functional F^p_σ is C^2 and satisfies Palais–Smale modulo reparametrization. If*

$$F^p_\sigma(u_k) \to \beta(\sigma) > 0, \qquad DF^p_\sigma(u_k) \to 0, \tag{16.3}$$

then, after a subsequence and diffeomorphisms φ_k,

$$u_k \circ \varphi_k \to v_\infty \quad \textit{strongly in } W^{2,2p}, \qquad DF^p_\sigma(v_\infty) = 0. \tag{16.4}$$

Critical points are smooth.

For an admissible family in $\mathcal{M}$, define

$$W_{\mathcal{A}}(\sigma) = \inf_{A\in\mathcal{A}} \sup_{\Phi\in A} F^p_\sigma(\Phi). \tag{16.5}$$

If $W_{\mathcal{A}}(\sigma) > 0$, Palais minmax yields a smooth critical immersion u_σ with

$$F^p_\sigma(u_\sigma) = W_{\mathcal{A}}(\sigma), \qquad DF^p_\sigma(u_\sigma) = 0. \tag{16.6}$$

16.1 Passing to $\sigma = 0$

Suppose $\sigma_k \to 0$, u_k are critical points, and

$$F^p_{\sigma_k}(u_k) \to W(0) > 0, \qquad \partial_\sigma F^p_{\sigma_k}(u_k) = o\left(\frac{1}{\sigma_k \log(1/\sigma_k)}\right). \tag{16.7}$$

Then there are a closed Riemannian surface (S, h), with $\mathrm{genus}(S) \le \mathrm{genus}(\Sigma)$, a weakly conformal map $v_\infty : S \to M$, and $N_\infty \in L^\infty(S, \mathbb{N})$ such that, after reparametrization,

$$(\Sigma, u_{k'}, 1) \rightharpoonup (S, v_\infty, N_\infty) \quad \text{as parametrized varifolds.} \tag{16.8}$$

The limiting object is locally stationary. The rest of the argument identifies its integer multiplicity and proves that its mass is the area width.

17 Varifolds and stationarity

Let $G_2(TM)$ denote the Grassmann bundle of unoriented tangent two-planes. For $U \in C^0(G_2(TM))$, the varifold associated with an immersion $u : \Sigma \to M$ is

$$v_u(U) = \int_\Sigma U\big(u(x), du(T_x\Sigma)\big)\ d\,\mathrm{vol}_{g_u}. \tag{17.1}$$

Equivalently,

$$dv_u(z, P) = d\big(\mathcal{H}^2 \llcorner u(\Sigma)\big) \otimes \sum_{u(x)=z} \delta_{du(T_x\Sigma)}. \tag{17.2}$$

Definition 17.1 (Integer rectifiable varifold)**.** Let $K \subset \mathbb{R}^n$ be countably 2-rectifiable and let $N : K \to \mathbb{N}^*$ be locally integrable with respect to $\mathcal{H}^2$. The integer rectifiable varifold $v = (K, N)$ is

$$dv(z, P) = N(z)\, d\mathcal{H}^2 \llcorner K(z) \otimes \delta_{T_z K}(P). \tag{17.3}$$

Its weight is $|v| = N\,\mathcal{H}^2 \llcorner K$, and its mass is $\|v\| = v(G_2(M))$.

For $\Phi \in \mathrm{Diff}(M)$, the push-forward is

$$\Phi_* v(U) = \int_{G_2(TM)} U(\Phi(z), d\Phi_z(P)) J\Phi(z, P)\, dv(z, P), \tag{17.4}$$

where $J\Phi(z, P)$ is the two-dimensional Jacobian. If Φ_t is the flow of a compactly supported vector field X, define the first variation

$$\delta v(X) = \frac{d}{dt}\|\Phi_{t*} v\|\Big|_{t=0}. \tag{17.5}$$

For an integer rectifiable varifold,

$$\delta v(X) = \int_K N(z)\, \mathrm{div}_{T_z K}\, X\, d\mathcal{H}^2(z), \tag{17.6}$$

where, for an orthonormal basis (e_1, e_2) of P,

$$\mathrm{div}_P\, X = \sum_{i=1}^{2} e_i \cdot \nabla^M_{e_i} X. \tag{17.7}$$

Definition 17.2. A varifold is stationary if $\delta v(X) = 0$ for every compactly supported vector field X.

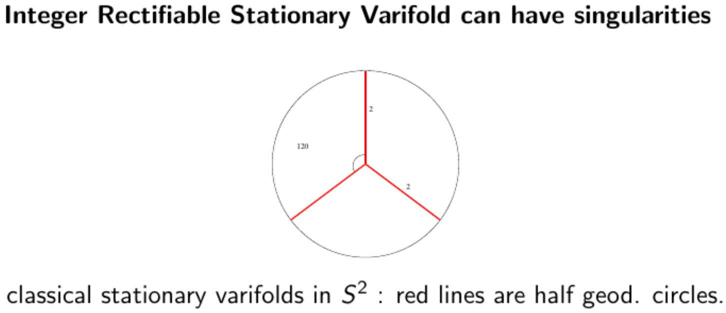

Figure 19: A stationary integer rectifiable varifold with a junction singularity.

Theorem 17.3 (Allard's weak closure)**.** *A weak Radon limit of stationary integer rectifiable varifolds is again stationary and integer rectifiable.*

Stationarity alone does not imply smoothness. Classical stationary varifolds on S^2 can have junctions and half-geodesic circles.

The special feature of the minmax limit is that it is parametrized by a weakly conformal map and carries an integer multiplicity. In local coordinates, if $u \in W^{1,2}(\Sigma, \mathbb{R}^n)$ is weakly conformal and $N \in L^\infty(\Sigma, \mathbb{N})$, the relevant stationarity identity is

$$\int_{\{f>t\}} N\, \nabla u \cdot \nabla[X(u)]\, dx^2 = 0 \tag{17.8}$$

for every compactly supported ambient vector field X and almost every regular level t of a domain test function f. The level set restriction supplies the missing variations in the naive question $\Delta u = 0$ from ambient variations alone.

Theorem 17.4 (Parametrized stationarity)**.** *If (Σ, u, N) satisfies the preceding identity, then N is essentially constant, u is smooth away from branch points, and u is a branched minimal immersion.*

17.1 Stationarity of the viscosity limit

Let $u_k = u_{\sigma_k}$. In a local conformal chart, criticality gives, for every ambient vector field X,

$$o(1) = \int_\Sigma du_k \cdot d(X \circ u_k)\; d\,\mathrm{vol}_{g_{u_k}} \tag{17.9}$$

$$= \int_{G_2(TM)} \mathrm{div}_P\, X(z)\, dv_k(z, P). \tag{17.10}$$

Weak convergence of v_k therefore implies

$$\int_{G_2(TM)} \mathrm{div}_P\, X(z)\, dv_\infty(z, P) = 0. \tag{17.11}$$

The candidate limit is

$$dv_\infty(z, P) = d\big(\mathcal{H}^2 \llcorner u_\infty(S)\big) \otimes \sum_{u_\infty(x)=z} N_\infty(x)\delta_{du_\infty(T_x S)}. \tag{17.12}$$

18 Monotonicity formulae

Consider first a weakly conformal harmonic map $u : \Sigma \to \mathbb{R}^m$. Testing $\Delta u = 0$ with $u\,\mathbf{1}_{u^{-1}(B_r(0))}$ and using the co-area formula yields

$$\frac{d}{dr}\left[\frac{1}{r^2}\int_{u^{-1}(B_r(0))} |\nabla u|^2\, dx^2\right] \geq 0. \tag{18.1}$$

By conformality, the energy is twice the area with multiplicity, so for every $q \in \mathbb{R}^m$

$$\frac{d}{dr}\left[\frac{\operatorname{Area}(u(\Sigma)\cap B_r(q))}{r^2}\right] \geq 0. \tag{18.2}$$

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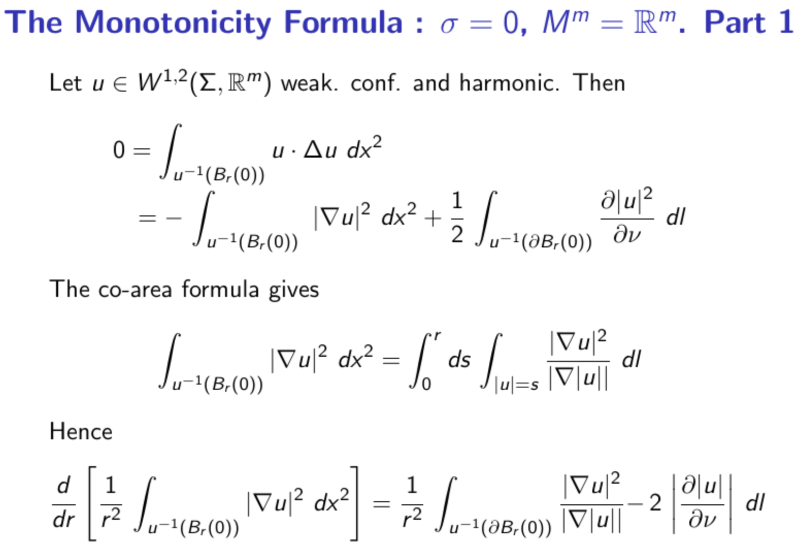

Figure 20: The unperturbed monotonicity formula for a conformal harmonic map.

For $\sigma \neq 0$, the Euler–Lagrange equation contains curvature errors. In $M = \mathbb{R}^3$, the perturbed formula has the schematic form

$$\frac{d}{dr}\left[e^{Cr}\frac{\operatorname{Area}(u(\Sigma)\cap B_r(q))}{r^2} - \frac{C_0\sigma^2}{r^2}\int_{\Sigma\cap B_r(q)} |d\vec{n}_u|^4 \; d\,\mathrm{vol}_{g_u}\right] \geq -\frac{C_0\sigma^2}{r^2}\int_{\Sigma\cap\partial B_r(q)} |d\vec{n}_u|^4 \; d\,\mathrm{vol}_{g_u}. \tag{18.3}$$

19 Energy quantization and the dichotomy lemma

The perturbed monotonicity formula replaces uniform ε-regularity.

> **Lemma 19.1** (Energy quantization)**.** *There is a universal $Q_0 > 0$ such that, whenever $\sigma_k \to 0$, u_k are critical points of*
>
> $$\operatorname{Area}(u) + \sigma_k^2 \int_\Sigma |d\vec{n}_u|^4 \; d\,\mathrm{vol}_{g_u}, \tag{19.1}$$

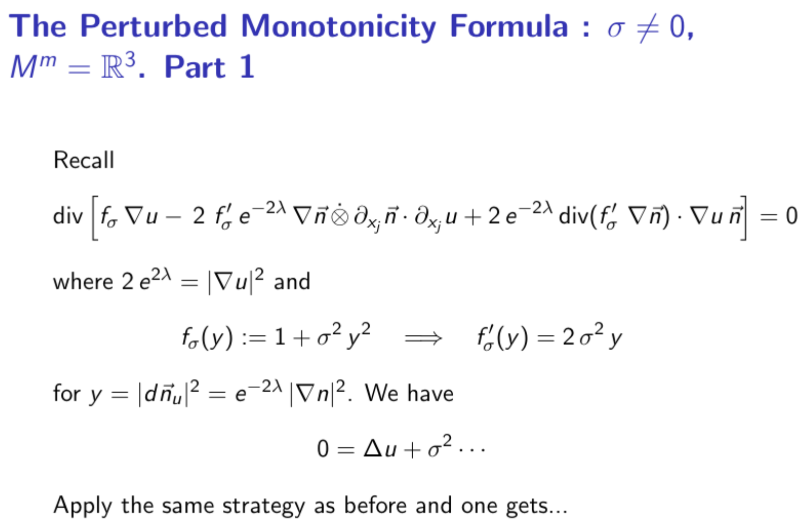

Figure 21: The perturbed monotonicity identity and its curvature error.

with bounded area and

$$\frac{\sigma_k^2 \int_\Sigma |d\vec{n}_{u_k}|^4 \; d\,\mathrm{vol}_{g_{u_k}}}{\mathrm{Area}(u_k)} = o\left(\frac{1}{\log(1/\sigma_k)}\right), \tag{19.2}$$

then

$$\liminf_{k\to\infty} \mathrm{Area}(u_k) \geq Q_0. \tag{19.3}$$

To see the mechanism, fix $\delta > 0$ and consider points q for which the scaled curvature energy is small on every ball of radius larger than σ, while the area at scale σ is not too small. A Besicovitch covering shows that the complement has vanishing area. Integrating the perturbed monotonicity formula between scales σ and one then gives a fixed positive amount of area near every remaining point. This yields Q_0.

Assume the conformal classes $[g_{u_k}]$ are precompact in moduli space and choose a representative h_∞ so that $[g_{u_k}] = h_\infty$. Define the energy measures

$$\nu_k = |du_k|^2_{h_\infty} \; d\,\mathrm{vol}_{h_\infty} \rightharpoonup \nu_\infty. \tag{19.4}$$

Lemma 19.2 (Dichotomy)**.** *There are $\varepsilon_0, C_0 > 0$ such that if $u_k \longrightarrow u_\infty$ uniformly on $\partial B_t(x_0)$ and*

$$s = \operatorname{diam} u_\infty(\partial B_t(x_0)) \leq \varepsilon_0, \tag{19.5}$$

then either

$$\nu_\infty(B_t(x_0)) \geq \frac{Q_0}{2}, \tag{19.6}$$

or

$$(u_k \llcorner B_t)_* \nu_k \rightharpoonup 0 \quad \text{outside } B_{4s}(u_\infty(x_0)), \tag{19.7}$$

$$\operatorname{diam} u_\infty(B_t(x_0)) \leq C_0 \operatorname{diam} u_\infty(\partial B_t(x_0)), \tag{19.8}$$

$$\sqrt{\nu_\infty(B_t(x_0))} \leq C_0 \operatorname{diam} u_\infty(\partial B_t(x_0)). \tag{19.9}$$

A substitute to the $\epsilon-$Regularity.

Energy Quantization Lemma There exists $Q_0 > 0$ s.t. for $\sigma_k \to 0$

- $$D\left[\text{Area} + \sigma_k^2 \int_\Sigma |d\vec{n}|^4 \, dvol_g\right](u_k) = 0 \ ,$$
- $$\limsup_{k\to+\infty} \text{Area}(u_k) < +\infty \ ,$$
- $$f(\sigma_k) := \frac{\sigma_k^2 \displaystyle\int_\Sigma |d\vec{n}_{u_k}|^4 \, dvol_{g_{u_k}}}{\text{Area}(u_k)} = o\left(\frac{1}{\log(1/\sigma_k)}\right) \ .$$

then

$$\liminf_{k\to+\infty} \text{Area}(u_k) \geq Q_0 > 0$$

□

Figure 22: Energy quantization is the substitute for uniform ε-regularity.

20 Continuity away from concentration points

The dichotomy lemma implies that only finitely many points can carry a mass at least $Q_0/2$. Denote them by $p_1, \dots, p_Q$.

Lemma 20.1. *There are points $p_1, \dots, p_Q \in \Sigma$ such that*

$$u_\infty \in C^0_{\text{loc}}(\Sigma \setminus \{p_1, \dots, p_Q\}). \tag{20.1}$$

Indeed, let x_0 satisfy $\nu_\infty(\{x_0\}) < Q_0/2$. Given $\varepsilon > 0$, choose $r > 0$ so that

$$\nu_\infty(B_{2r}(x_0)) < Q_0/2, \qquad \int_{B_{2r}(x_0)} |\nabla u_\infty|^2 \, dx^2 < \varepsilon^2. \tag{20.2}$$

For some $t \in (r, 2r)$, co-area and weak convergence give

$$\operatorname{diam} u_\infty(\partial B_t(x_0)) \leq \int_{\partial B_t(x_0)} |\nabla u_\infty| \, dl \leq 4 \left(\int_{B_{2r}(x_0)} |\nabla u_\infty|^2 \, dx^2\right)^{1/2}, \tag{20.3}$$

$$\limsup_k \int_{\partial B_t(x_0)} |\nabla u_k|^2, dl \leq r^{-1} \nu_k(B_{2r}(x_0)). \tag{20.4}$$

Thus $u_k \to u_\infty$ uniformly on the circle, and the dichotomy lemma gives

$$\operatorname{diam} u_\infty(B_t(x_0)) \leq C\varepsilon. \tag{20.5}$$

This proves continuity at every point outside the finite atom set.

21 The structure of the limiting measure

21.1 The preliminary decomposition

Lemma 21.1. *There are $f \in L^1(\Sigma)$ and positive constants c_i such that*

$$\nu_\infty = f \ d\,\text{vol}_{h_\infty} + \sum_{i=1}^{Q} c_i \, \delta_{p_i}. \tag{21.1}$$

To prove this, cover a compact set $K \subset \Sigma \setminus \{p_1, \dots, p_Q\}$ of zero $\mathcal{H}^2$-measure by balls $B_r(x_i)$ with bounded overlap of the doubled balls. Choosing $t_i \in (r, 2r)$, the dichotomy estimate gives

$$\nu_\infty(B_r(x_i)) leq C_0 \left(\int_{B_{2r}(x_i)} |\nabla u_\infty|^2 \, dx^2 \right). \tag{21.2}$$

Summation yields

$$\nu_\infty(K) leq C \int_{\{x:\mathrm{dist}(x,K)<2r\}} |\nabla u_\infty|^2 \, dx^2, \tag{21.3}$$

which tends to zero as $r \to 0$. The non-atomic part is therefore absolutely continuous with respect to $d\,\mathrm{vol}_{h_\infty}$.

21.2 Convergence relative to ν_k

Lemma 21.2. *For every compact $K \subset \Sigma \setminus \{p_1, \dots, p_Q\}$,*

$$\int_K |u_k - u_\infty| \, d\nu_k \longrightarrow 0, \tag{21.4}$$

and

$$(u_k \llcorner K)_* \nu_k \rightharpoonup (u_\infty \llcorner K)_* \nu_\infty. \tag{21.5}$$

Cover K by finitely many small balls with controlled overlap and choose boundary circles on which $u_k \to u_\infty$ uniformly. If s_i denotes the diameter of the limiting image of a chosen circle, the dichotomy lemma gives

$$\int_{B_{t_i}} |u_k - u_\infty| \, d\nu_k \le C s_i \nu_k(B_{t_i}) + o(1). \tag{21.6}$$

Taking the maximum over the covering and then letting its radius tend to zero proves the lemma.

21.3 Integer multiplicity

Theorem 21.3 (Final structure of ν_∞)**.** *There are $N \in L^\infty(\Sigma, \mathbb{N})$ and positive constants c_i such that*

$$\nu_\infty = N \, |du_\infty \wedge du_\infty|_{h_\infty} \, d\,\mathrm{vol}_{h_\infty} + \sum_{i=1}^{Q} c_i \, \delta_{p_i}. \tag{21.7}$$

In local coordinates,

$$|du_\infty \wedge du_\infty|_{h_\infty} \, d\,\mathrm{vol}_{h_\infty} = |\partial_{x_1} u_\infty \wedge \partial_{x_2} u_\infty| \, dx_1 \, dx_2. \tag{21.8}$$

We record the blow-up argument, since it explains why the density is integer-valued. Choose a Lebesgue point p for f, u_∞, and conformal coordinates with

$$h_\infty = e^{2\lambda}(dx_1^2 + dx_2^2), \qquad x(p) = 0, \qquad f(0) > 0. \tag{21.9}$$

For radii $r \to 0$, differentiation and a diagonal extraction give $r_j \to 0$, $k_j \to \infty$ such that

$$|\nu_{k_j}(B_{r_j}) - \nu_\infty(B_{r_j})| = o(r_j^2), \tag{21.10}$$

$$\fint_{B_{r_j}} |u_{k_j} - u_\infty|^2 \, dx^2 = o(r_j^2), \tag{21.11}$$

$$\fint_{B_{r_j}} |u_{k_j} - u_\infty| \, d\nu_{k_j} = o(r_j), \tag{21.12}$$

$$\frac{\sigma_{k_j}}{r_j^2} = o(1). \tag{21.13}$$

Rescale by

$$x = r_j y, \qquad \widehat{u}_j(y) = r_j^{-1} u_{k_j}(r_j y), \qquad \widehat{\vec{n}}_j(y) = \vec{n}_{k_j}(r_j y), \tag{21.14}$$

and set $\widehat{\sigma}_j = \sigma_{k_j}/r_j^2$. Then $\widehat{u}_j$ is critical for

$$\widehat{F}_j(\widehat{u}) = \int_{B_1} d_{\widehat{u}} + \widehat{\sigma}_j^2 \int_{B_1} |d\widehat{\vec{n}}|^4 \, d_{\widehat{u}}, \tag{21.15}$$

with

$$\widehat{F}_j(\widehat{u}_j) \longrightarrow \pi f(0) e^{2\lambda(0)}, \qquad \widehat{\sigma}_j^2 \int_{B_1} |d\widehat{\vec{n}}_j|^4 \, d_{\widehat{u}_j} = o\left(\frac{1}{\log(1/\widehat{\sigma}_j)}\right). \tag{21.16}$$

The rescaled limit is affine:

$$\widehat{u}_{\infty,\infty}(y) = \partial_{x_1} u_\infty(0) y_1 + \partial_{x_2} u_\infty(0) y_2. \tag{21.17}$$

After a rotation, its image plane is

$$\mathcal{P}_0 = \mathrm{Span}\{(1, 0, \dots, 0), (0, 1, 0, \dots, 0)\}. \tag{21.18}$$

The transverse components of $\widehat{u}_j$ vanish in the limit, so the rescaled varifolds converge to a measure supported on $\mathcal{P}_0$. Stationarity and the constancy theorem imply

$$d\widehat{\nu}_\infty = \theta_0 \, d\mathcal{H}^2 \llcorner \mathcal{P}_0 \otimes \delta_{\mathcal{P}_0} \tag{21.19}$$

for some $\theta_0 > 0$.

Let $\widetilde{u}_j = (\widehat{u}_j^1, \widehat{u}_j^2)$. Testing the equation with smooth functions of $\widetilde{u}_j$, using almost conformality and the co-area formula, shows that the counting function

$$N_j(z) = \mathrm{Card}\big(\widetilde{u}_j^{-1}(z)\big) \tag{21.20}$$

has vanishing distributional gradient in the interior. The weak L^1 estimate implies

$$\left\| N_j - \fint_{B_{1/2}} N_j \right\|_{L^{1,\infty}(B_{1/2})} \longrightarrow 0. \tag{21.21}$$

Since N_j is integer-valued, it converges to a constant $N_\infty \in \mathbb{N}^*$. Finally, the area formula gives

$$\pi f(0) e^{2\lambda(0)} = \lim_{j\to\infty} \int_{B_1} \frac{1}{2} |\nabla \widehat{u}_j|^2 \, dy^2 \tag{21.22}$$

$$= \int_{\widehat{u}_{\infty,\infty}(B_1)} N_\infty \, dp^2 \tag{21.23}$$

$$= \pi N_\infty |\partial_{x_1} u_\infty(0) \wedge \partial_{x_2} u_\infty(0)|. \tag{21.24}$$

Thus $f = N_\infty |du_\infty \wedge du_\infty|$ almost everywhere.

22 Conclusion of the minmax construction

The limiting parametrized varifold is stationary, integer rectifiable, and represented by the smooth minimal branched immersion $v_\infty : S \to M$ with multiplicity N_∞. Lower semicontinuity and the definition of the viscous widths give

$$\|v_\infty\| = \lim_{k\to\infty} F^p_{\sigma_k}(u_k) = W_{\mathcal{A}}. \tag{22.1}$$

The parameter family controls the Morse index by the dimension of the homological sweep-out, and the conformal compactness argument gives $\mathrm{genus}(S) \leq \mathrm{genus}(\Sigma)$. This completes the minmax realization of the area by a minimal branched immersion.

A Notation and conventions

All surfaces are smooth, closed and oriented unless stated otherwise. The symbol d_g denotes the Riemannian volume form, while $d\,\mathrm{vol}_{g_u}$ is the area measure induced by an immersion. Weak convergence of measures is denoted by $\rightharpoonup$. The notation $\mathcal{H}^2 \llcorner K$ means restriction of Hausdorff measure to K. Constants denoted by C may change from line to line but depend only on the fixed geometric data and the displayed bounds.